Isotropic noises do not force an isotropic Riccati covariance

Disproof of Conjecture 00000008262

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Abstract

The conjectured bound for the spectral condition number of a steady Kalman–Bucy covariance fails already in two dimensions with equal, nondegenerate isotropic system and observation noises. The stabilizing covariance has eigenvalues 2 and 1/2, hence condition number 4, while the proposed upper bound is 1.

The claimed bound and the filtering model

Both language versions assert

$$\kappa_2(P_\infty) \le (\sigma_{\mathrm{obs}}/\sigma_{\mathrm{sys}})^2.$$

We refute this first conjunct. No conclusion about the other claimed transient-rate and robustness-radius formulas is needed.

Consider the usual continuous-time linear filtering model

$$dX_t = AX_t\,dt + \sigma_{\mathrm{sys}}\,dW_t, \qquad dY_t = HX_t\,dt + \sigma_{\mathrm{obs}}\,dV_t,$$

with independent standard two-dimensional Brownian motions. The covariance Riccati equation and its steady equation are

$$\begin{aligned}\dot P &= AP + PA^T + \sigma_{\mathrm{sys}}^2 I - \sigma_{\mathrm{obs}}^{-2} PH^T HP,\\ 0 &= AP_\infty + P_\infty A^T + \sigma_{\mathrm{sys}}^2 I - \sigma_{\mathrm{obs}}^{-2} P_\infty H^T H P_\infty.\end{aligned}$$

Set

$$A = \begin{pmatrix} 3/4 & 0 \\ 0 & -3/4 \end{pmatrix}, \quad H = I, \quad \sigma_{\mathrm{sys}} = \sigma_{\mathrm{obs}} = 1, \quad P_\infty = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

All noise covariances are strictly positive definite. In particular, the example does not use a zero-noise or unobservable direction.

Verification of the steady covariance

The matrix P_∞ is real symmetric positive definite. Direct multiplication gives

$$AP_\infty + P_\infty A^T = \mathrm{diag}(3, -3/4), \qquad P_\infty^2 = \mathrm{diag}(4, 1/4),$$

so $AP_\infty + P_\infty A^T + I - P_\infty^2 = 0$. The closed-loop error matrix is

$$A - P_\infty H^T H = A - P_\infty = -\frac{5}{4} I.$$

Both complex eigenvalues therefore have strictly negative real part; this is a stabilizing positive-definite Riccati solution.

One can also see the actual steady limit without invoking a general Riccati existence theorem. With $P(0) = 0$, put $e(t) = \exp(-5t/2)$ and

$$P(t) = \operatorname{diag}\left(\frac{2(1-e(t))}{1+4e(t)}, \frac{(1/2)(1-e(t))}{1+e(t)/4}\right).$$

Differentiation verifies the covariance equation, and the positive denominators show it is defined for all $t \geq 0$. It converges to the displayed P_∞. Alternatively, initializing the covariance at P_∞ gives an exactly stationary solution. These observations ensure that the example is a genuine steady filtering covariance, not an unrelated algebraic root.

Contradiction

For a real symmetric positive-definite matrix its spectral condition number is the ratio of its largest to its smallest eigenvalue. Here the real spectrum is exactly $\{2, 1/2\}$, and hence

$$\kappa_2(P_\infty) = \frac{2}{1/2} = 4 > 1 = (\sigma_{\text{obs}}/\sigma_{\text{sys}})^2.$$

The same numerical contradiction holds if the symbols for the noises are interpreted as variances rather than standard deviations, since both equal one. The conjecture has no hypothesis excluding anisotropic system dynamics A. Such dynamics are precisely what its proposed noise-only condition-number bound overlooks.

Formal verification and scope

Lean uses actual Mathlib matrices, `Matrix.PosDef`, and the standard real and complex matrix spectra. The predicate `RiccatiSteady` is the full displayed algebraic matrix equation with arbitrary matrices A, H, P and positive scalar noise amplitudes. The predicate `Hurwitz` quantifies over the complex spectrum of the complexified closed-loop matrix and requires negative real parts. The predicate `SpectralCondition` defines the spectral condition number using the least and greatest elements of the real spectrum and their ratio, with strictly positive least element.

The development proves positive definiteness, the actual Riccati equation, the closed-loop stability, the exact real spectrum and condition number, and negates the universally quantified proposed bound. Its final theorem is

`Conjecture8262.conjecture8262_false`.

There is no table substituted for matrix operations or spectra. The complex and real spectra of the diagonal matrices are obtained from Mathlib's general diagonal-spectrum theorem.

The Brownian filtering model and the explicit time-dependent solution above are justified on paper. Lean does not formalize stochastic conditional covariance, the covariance ODE, or the general Kalman–Bucy theorem. The formal result refutes the claimed bound directly on stabilizing positive-definite solutions of its actual steady Riccati equation. These are the steady covariances in this fully observable, nondegenerate example, as the explicit solution independently checks.

Source and reproduction

The unmodified bilingual source is supplied as `SOURCE.md`, with frozen-source provenance and SHA-256 in the verification folder. The parent workflow found no previous submission for this conjecture. Lean is pinned to version 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

Run in the supplied Lean folder:

```
lake env lean -DwarningAsError=true Main.lean
```

The printed axiom audits use only `propext`, `Classical.choice`, and `Quot.sound`. This draft was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.